



KOTLIN USER GROUP BERLIN

One Topic to Rule 'Em All

Taming domain-event order with Kafka & Avro

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github.com/yonatankarp/one-topic-to-rule-em-all



Adventurer: Yonatan Karp-Rudin

Class: Staff Engineer · **Guild:** Billie (Berlin)

Weapons of choice: Kotlin, Spring Boot, and event-driven systems that occasionally fight back

Inventory: `ff4k` · `kotlin-design-patterns` · `kotlin-spring-boot-template`

Current quest: *making domain events arrive in the order they happened*



Yonatan

Something just leveled up a hero who doesn't exist

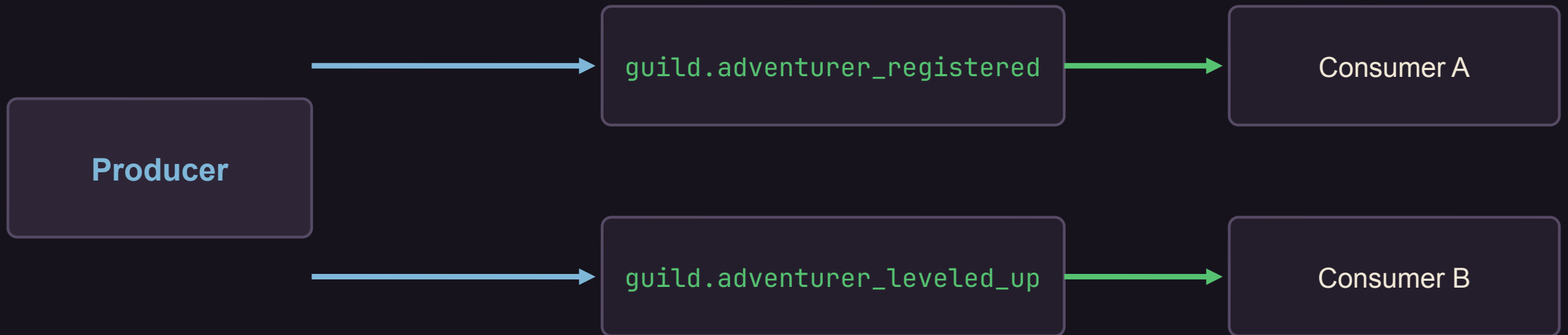
✦ LEVEL-UP FOR UNKNOWN ADVENTURER ad-7f3c1 -
the ledger has never heard of them!

Every event was delivered. Exactly once. In the order you produced them.

Every CRUD shop has this bug - wearing a different costume.

THE SETUP EVERYONE BUILDS

One topic per event type. Looks clean. Ships fast.

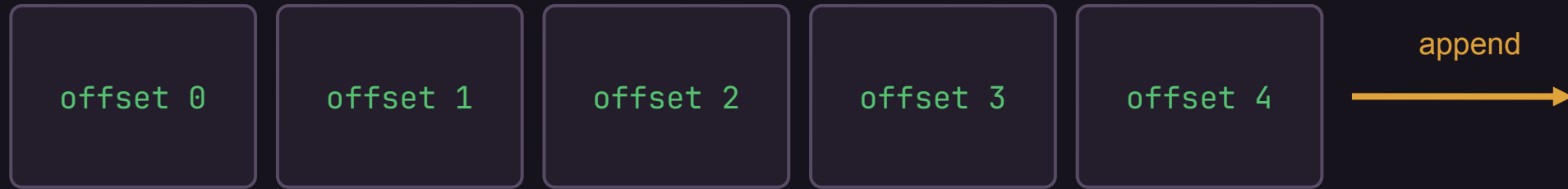


Type-named topics. Clear ownership. Easy diagram. Approved in review.

WHAT KAFKA ACTUALLY GUARANTEES

Ordering exists within a partition. Full stop.

partition 0



Producer writes in order → consumer reads in that exact order.

The partition is the unit of order. Nothing larger is ordered.

COROLLARY

Across topics, there is NO ordering. None.

`guild.adventurer_registered`

lag: high (busy consumer)

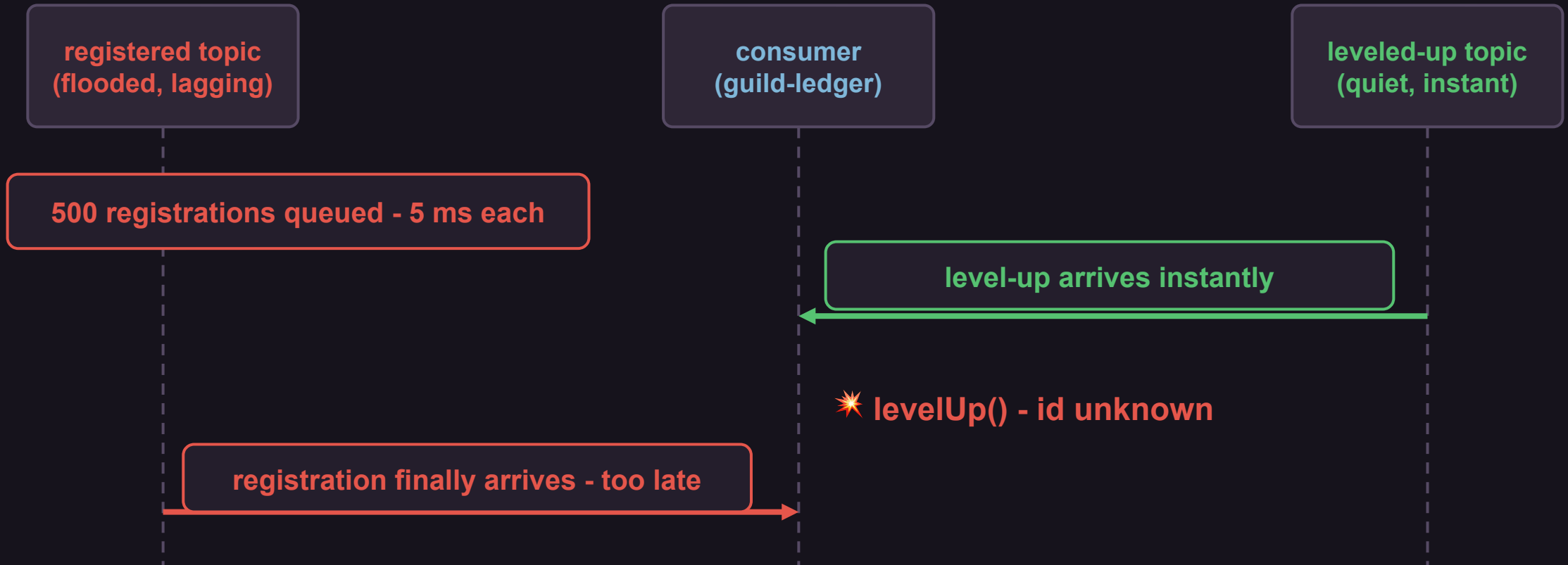
`guild.adventurer leveled_up`

lag: ~zero (idle consumer)

Two topics = two independent clocks. They drift.

Star Wars shipped IV-VI before I-III. Your topics do this daily.

How the level-up overtakes the registration



THE TEMPTING NON-FIX

"Can't I just buffer and reorder by timestamp?"

- How long do you wait for the maybe-late event?
- Whose clock? Producer skew, NTP drift, GC pauses.
- What if it never comes - is it lost, or just slow?
- **You now own a stateful reordering buffer and its bugs.**

You've rebuilt a partition. Badly.

THE FIX

All events of one aggregate go to **one topic, keyed by the **aggregate id**.**

Aggregate: the thing whose events must stay in order.

Same key → same partition → guaranteed order. That's the whole trick.

Where each event lands

```
// BROKEN: one topic per type
override fun topicFor(e: SpecificRecord) = when (e) {
    is AdventurerRegistered → "guild.adventurer_registered"
    is AdventurerLeveledUp   → "guild.adventurer_leveled_up"
    else → error("no topic for ${e.schema.name}")
}
```

```
// FIXED: one aggregate, one topic
override fun topicFor(e: SpecificRecord) = "guild.adventurers"
```

The only producer-side change between the acts. That broken else is where QuestAccepted and Died vanish.

BUT THE REGISTRY SAYS NO

Register a 2nd schema on guild.adventurers-value

```
POST /subjects/guild.adventurers-value/versions
{ schema: AdventurerLeveledUp }
```

409 Conflict

Schema being registered is incompatible with
an earlier schema for subject
"guild.adventurers-value"

Default subject naming assumes ONE schema per topic. A 2nd type looks like a breaking change.

A subject is not a topic

- Topic - Kafka's storage & ordering unit (the log).
- Subject - the registry's versioned compatibility scope.
- **Compatibility is checked per subject - not per topic.**
- **topic ↔ schema as 1:1 is just the DEFAULT naming strategy.**

Change the naming strategy and one topic can carry many independently-versioned schemas.

THE THREE SUBJECT-NAMING STRATEGIES

Pick the subject pattern, change everything

TopicNameStrategy

`<topic>-value`

`guild.adventurer_registered-value`

The default - and the cage.

RecordNameStrategy

`<record FQN>`

`org.guild.adventurer.AdventurerRegistered`

Schema roams any topic; niche.

TopicRecordNameStrategy

`<topic>-<record FQN>`

`guild.adventurers-`

`org.guild.adventurer.AdventurerRegistered`

Many schemas, scoped per topic.

WINNER

THE ONE LINE

application-fixed.yml

```
spring:
  kafka:
    producer:
      properties:
        # The one line this talk is about:
        value.subject.name.strategy:
          io.confluent.kafka.serializers.subject.TopicRecordNameStrategy
```

Producer-side only. Consumers read the schema id from the record bytes - no consumer config change.

Avro IDL - AdventurerRegistered.avdl

```
@namespace("org.guild.adventurer")
protocol AdventurerRegisteredProtocol {
  /** A new adventurer signs the guild ledger. High-volume event. */
  record AdventurerRegistered {
    string adventurer_id;
    string name;
    string character_class;
    timestamp_ms registered_at;
  }
}
```

Four lifecycle events. One hero.

- ▶ **AdventurerRegistered** *the flood - 500/run*
- ▲ **AdventurerLeveledUp** *rare - the overtaker*
- ◆ **QuestAccepted** *occasional*
- ✦ **AdventurerDied** *the finale*



Bob the Brave

Level 1 Fighter - our hero

Bob runs this exact lifecycle, top to bottom. Watch for the failure line - spot the bug before the consumer does.

◆ LIVE DEMO

Break it. Then fix it.

Act 1 two topics - the level-up overtakes the registration → ✨ **unknown adventurer**

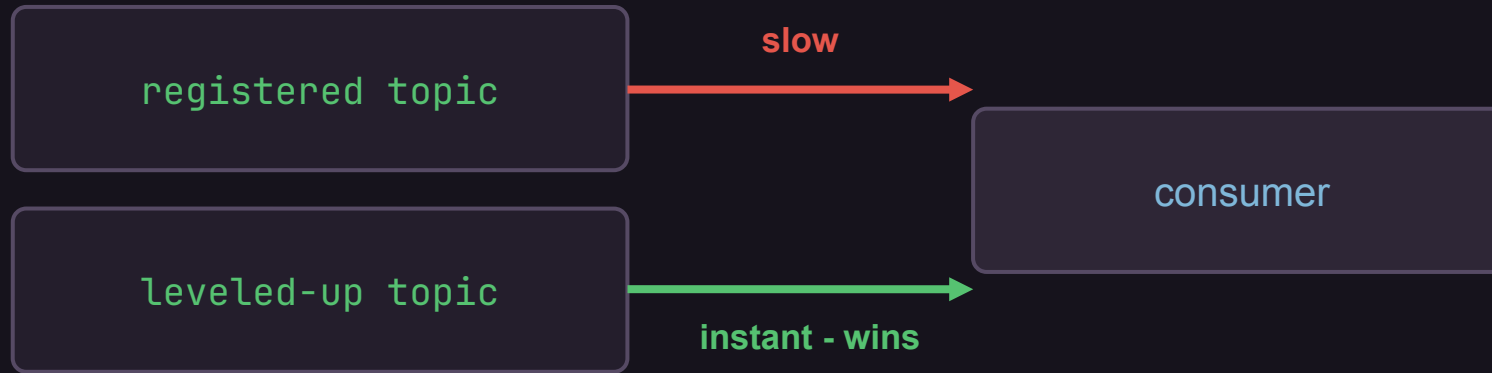
Act 2 one topic, one line - same flood, same hero, order holds → 🍺 **tavern fan-out**

Same scenario both times. Only the producer router + one yml line differ.

→ to the terminals

WHAT YOU JUST SAW

The overtake, and the receipts



- ✦ LEVEL-UP FOR UNKNOWN ADVENTURER ad-7f3c1 - the ledger has never heard of them!
- QuestAccepted has nowhere to go in this topology

Not a Kafka bug. Kafka did exactly what we asked: two topics, two clocks.

The class reads like the aggregate's lifecycle

```
@KafkaListener(topics = ["guild.adventurers"], groupId = "guild-ledger")
class GuildLedgerListener(/* ... */) {
    @KafkaHandler fun on(event: AdventurerRegistered) { ... }
    @KafkaHandler fun on(event: AdventurerLeveledUp)   { ... }
    @KafkaHandler fun on(event: QuestAccepted)         { ... }
    @KafkaHandler fun on(event: AdventurerDied)        { ... }
    @KafkaHandler(isDefault = true)
    fun unknown(event: Any) {
        log.warn("Unhandled event type: {}", event::class.simpleName)
    }
}
```

@KafkaHandler vs. when - same hole

@KafkaHandler - framework dispatch

- One method per type.
- Spring routes by deserialized type.
- Forgotten type → silent isDefault, at runtime.

when dispatch - one explicit place

- All routing in one greppable when.
- Avro classes aren't sealed → else required.
- Forgotten type → falls to else, at runtime.

Neither gives compile-time safety on raw Avro types. For that, map events into your own sealed hierarchy at the boundary.

Both variants live in the repo - profiles

`fixed` / `fixed-when`.

Make the topology obvious, let CI keep it honest

- Topic = domain.aggregate; the name IS the consistency boundary.
- **TopicRecordNameStrategy** → each event type is its own subject, versioned on its own.
- So you evolve one event type without touching its siblings.
- Schemas live in a central repo, reviewed like code; CI gates compatibility per subject.
- **Producers run auto.register.schemas = false** → a break fails the build, not 3 a.m. in prod.

Naming makes the boundary readable; CI makes compatibility a build-time gate, not a runtime surprise.

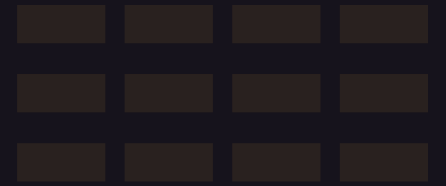
Every consumer sees every type on the topic

- All consumers deserialize all types landing on the topic.
- Filtering is consumer-side (e.g. `@KafkaHandler` / `when`).
- **Fine for aggregate streams - types are few and related.**
- **Wrong for firehose topics - many unrelated, high-volume types.**

Topic-per-event-type is correct when...

- The streams are genuinely independent - no shared aggregate.
- No ordering relationship to protect between the events.
- Heterogeneous retention or scaling per type.
- **No single aggregate id ties the records together.**

The rule isn't 'one topic always.' Split by aggregate boundary, not by event type.



Four things to take home

- 1 Ordering is a partition guarantee - and nothing larger.
- 2 One aggregate → one topic, key = aggregate id.
- 3 TopicRecordNameStrategy is the knob that unlocks it.
- 4 @KafkaHandler makes the multi-type topic pleasant.

 **Thank you!**

Questions?

The full runnable demo - broken + fixed - lives in the repo.

github.com/yonatankarp/one-topic-to-rule-em-all

FIND ME

github.com/yonatankarp · linkedin.com/in/yonatankarp
medium.com/@yonatankarp · yonatankarp.com



scan for the repo



APPENDIX

Backup slides

Schema evolution · union schemas

Evolve one event without touching its siblings

- Compatibility is checked per subject = per (topic, event type).
- Add a field to AdventurerRegistered → only that subject evolves.
- **AdventurerLeveledUp, QuestAccepted, Died are untouched.**
- **Independent version histories per type on the same topic.**

Union schema + schema references

- Model the topic value as one union of all event types (via schema refs).
- Single subject, single schema id namespace for the topic.
- Trade-off: atomic evolution of the whole topic shape...
- **...but you lose per-type compatibility isolation.**

TopicRecordNameStrategy = many subjects, isolated. Union = one subject, atomic. Pick your coupling.